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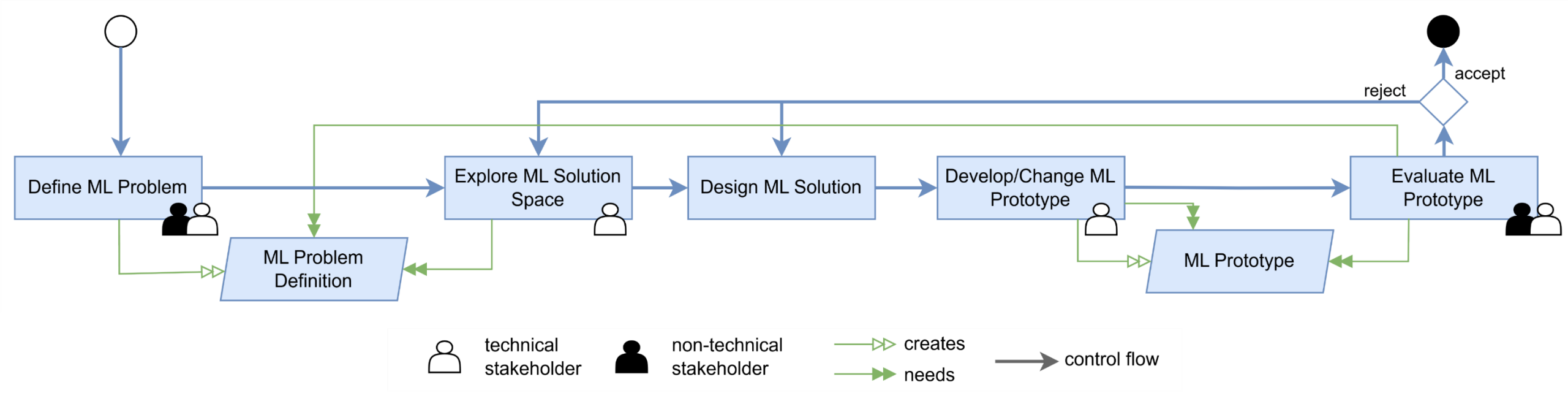
AI supported Information Foraging

Kickoff Software Lab Winter Semester 2025/2026

SWC Software Construction
RWTH AACHEN UNIVERSITY

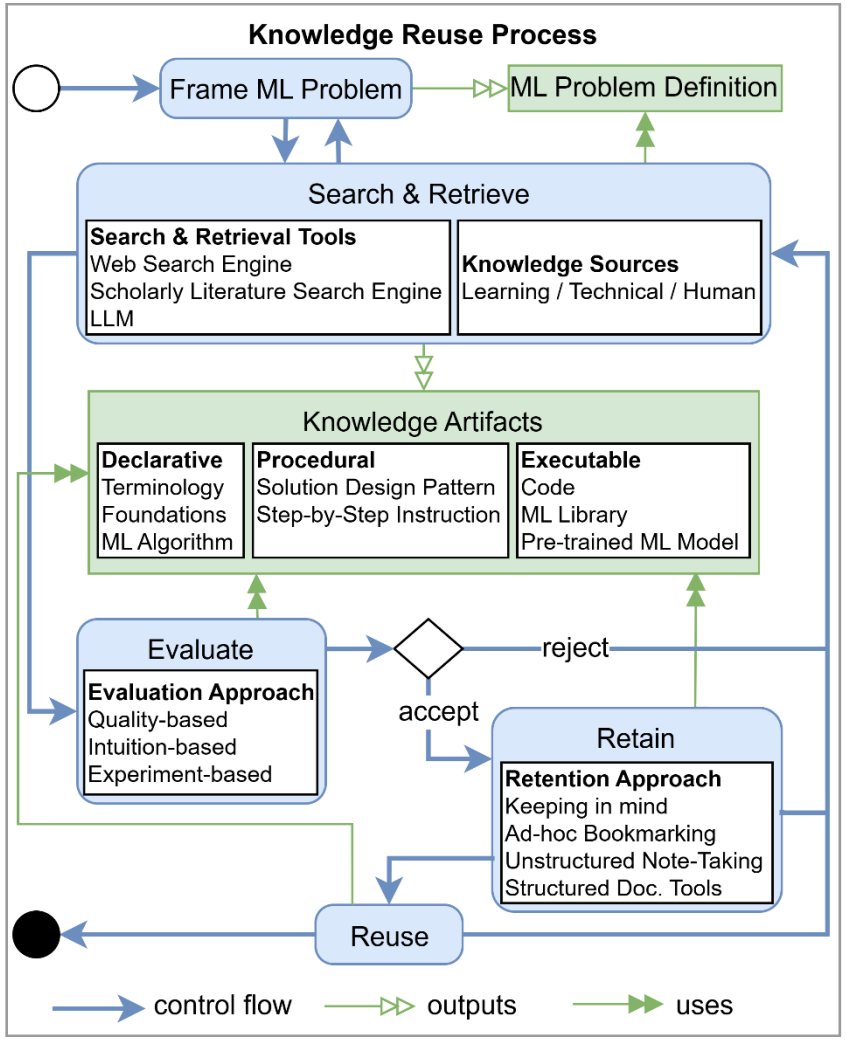
Lab Task

ML Solution Prototyping

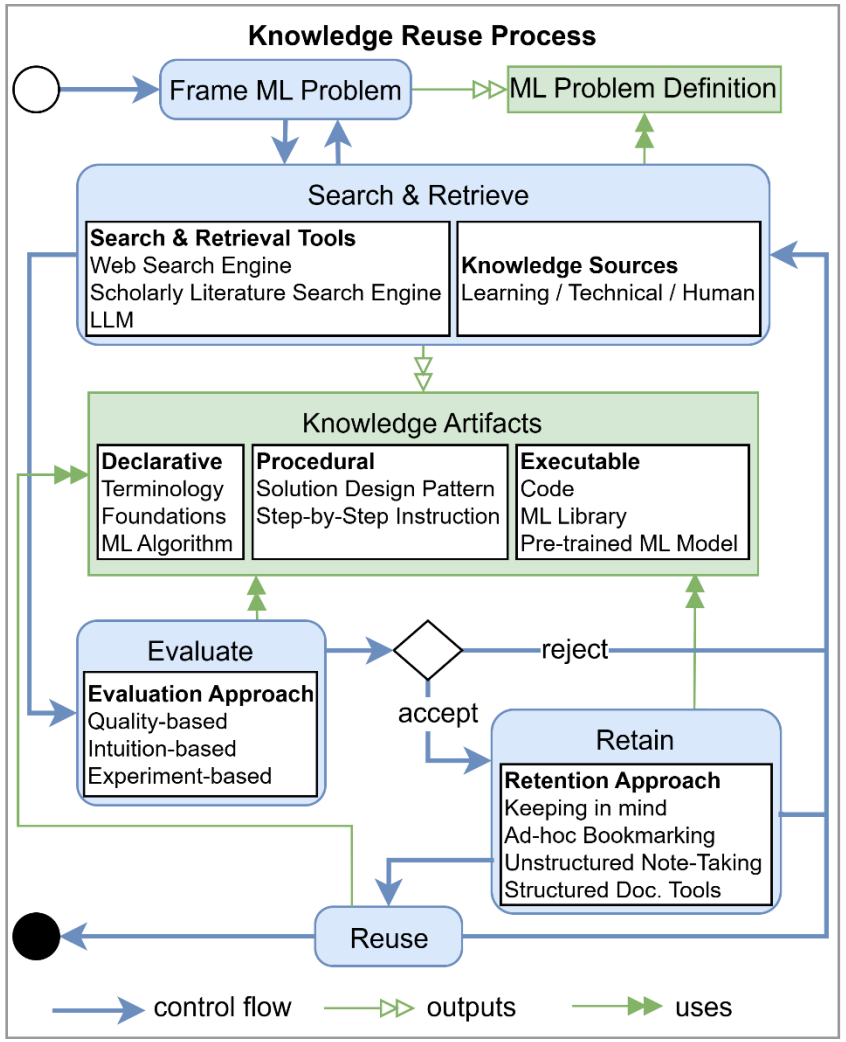


Reuse in ML Solution Prototyping

Interview Study with 18 participants from academia/industry

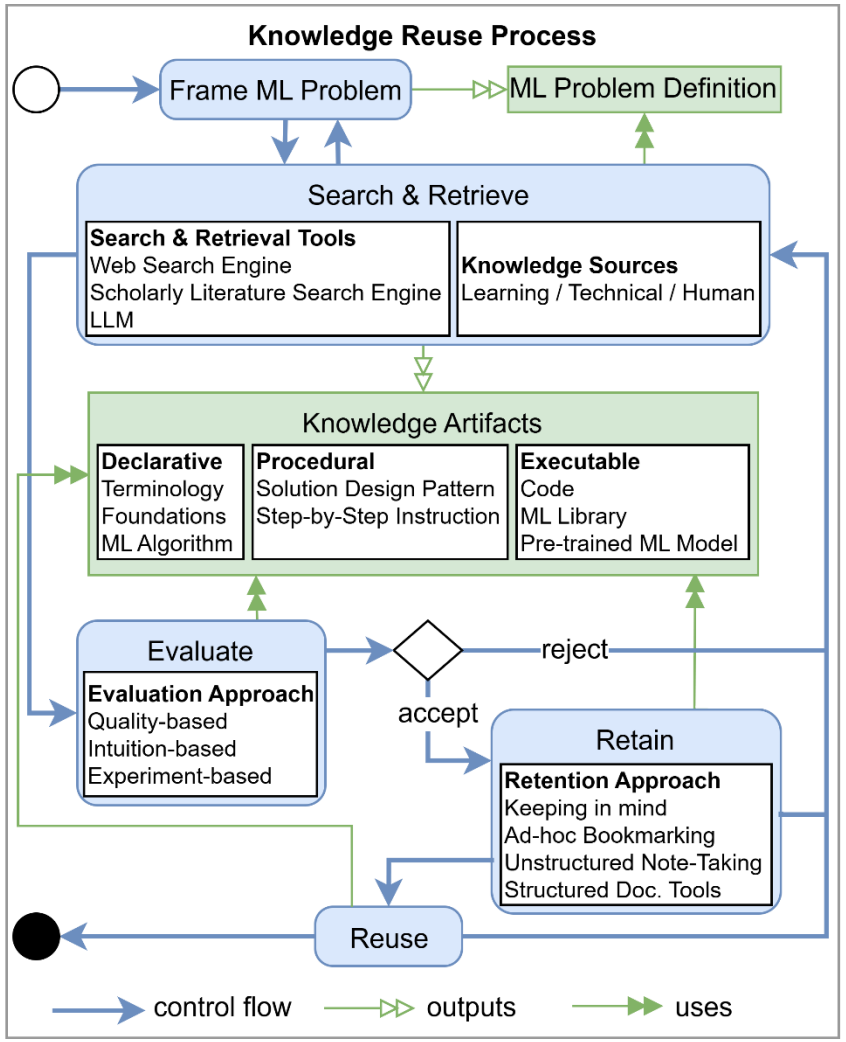


Information Foraging in ML Solution Prototyping



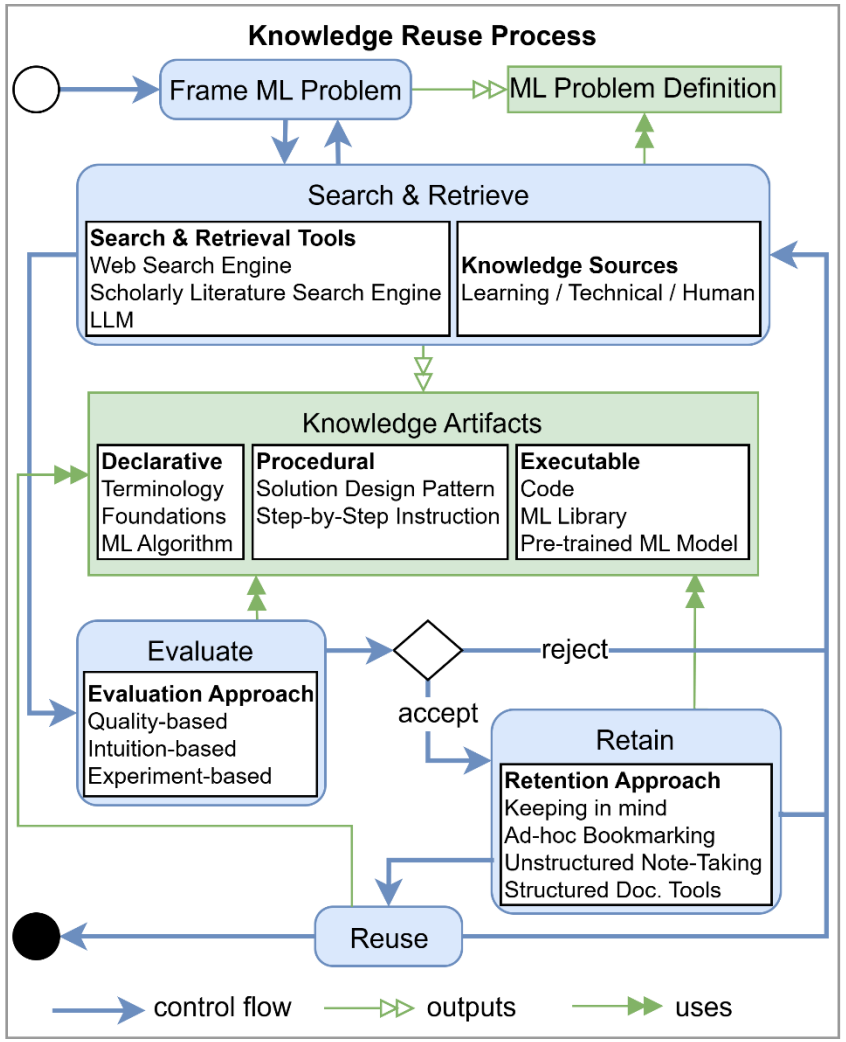
Category	Knowledge Source Type	#Part.
Sources for Learning	Blog Article	16
	Scientific Paper	11
	Video	10
	Book	6
Sources for Coding	Stack Overflow Post	14
	Technical Documentation	10
	Personal Code Archive	10
	Git Repository	9
Human Sources	Colleague / Expert	17

Information Foraging in ML Solution Prototyping



Quality	Quality Criterion	#Part.
Usability	Compatibility with development environment	15
	Writing style	14
	Ease of understanding	14
	Presence of examples, code, or tutorials	12
	Effort required to reuse the solution	10
Credibility	Recognized author or publisher	14
	Community feedback	12
	Transparency of decisions and results	11
	Match with personal knowledge or experience	8
	Cited or mentioned frequently	6
Relevance	Context match	17
	Recency of publication or update	14

Information Foraging in ML Solution Prototyping



Retention Approach	Mentioned Example	#Part.
Absent	Rely on memory only	10
Ad hoc	Keeping tabs open	7
Bookmarking	Revisiting search history	2
Unstructured	Taking digital or physical notes	8
Note taking	Leaving comments in code	3
Structured	Notes in dedicated apps	6
Documentation	Curating collection of links	5
Tools		

Task

- This lab is part of an active research project
  - Successful results could lead to a publication featuring you as a co-author!

- Build a knowledge manager that allows to
  - Manage different knowledge sources (paper, blog articles, etc.)
  - Annotate with relevant information (e.g. criteria from quality model)
  - Support automatic evaluation of knowledge sources
  - Trace parts from knowledge sources to ML prototype parts (annotation mechanism?)

Challenges

- Manage different knowledge sources (paper, blog articles, etc.)
  - How to model knowledge sources and elements?
  - How to manage? Are there existing tools to build on (e.g. bibtex, JabRef?)?
- Annotate with relevant information (e.g. criteria from quality model)
  - Define relevant information!
- Support automatic evaluation of knowledge sources
  - Suitable AI method?
- Trace parts from knowledge sources to ML prototype parts (annotation mechanism?)
  - Integration with development tools and workflow?
- Integration of software and AI components!

Lab Organization

Organization

- In-person Lab Sessions (attendance mandatory*!):
  - Tuesdays, 10:30 Uhr in 5054 (?)
- Zoom Weekly: tbd

- If not interested: one week to deregister

- Midterm presentation of work done around December
- Final presentation in last lecture week

- Workload: 7 ECTS = 210h = 2,5h per working day

*suspended in justified cases, e.g. illness

Technologies

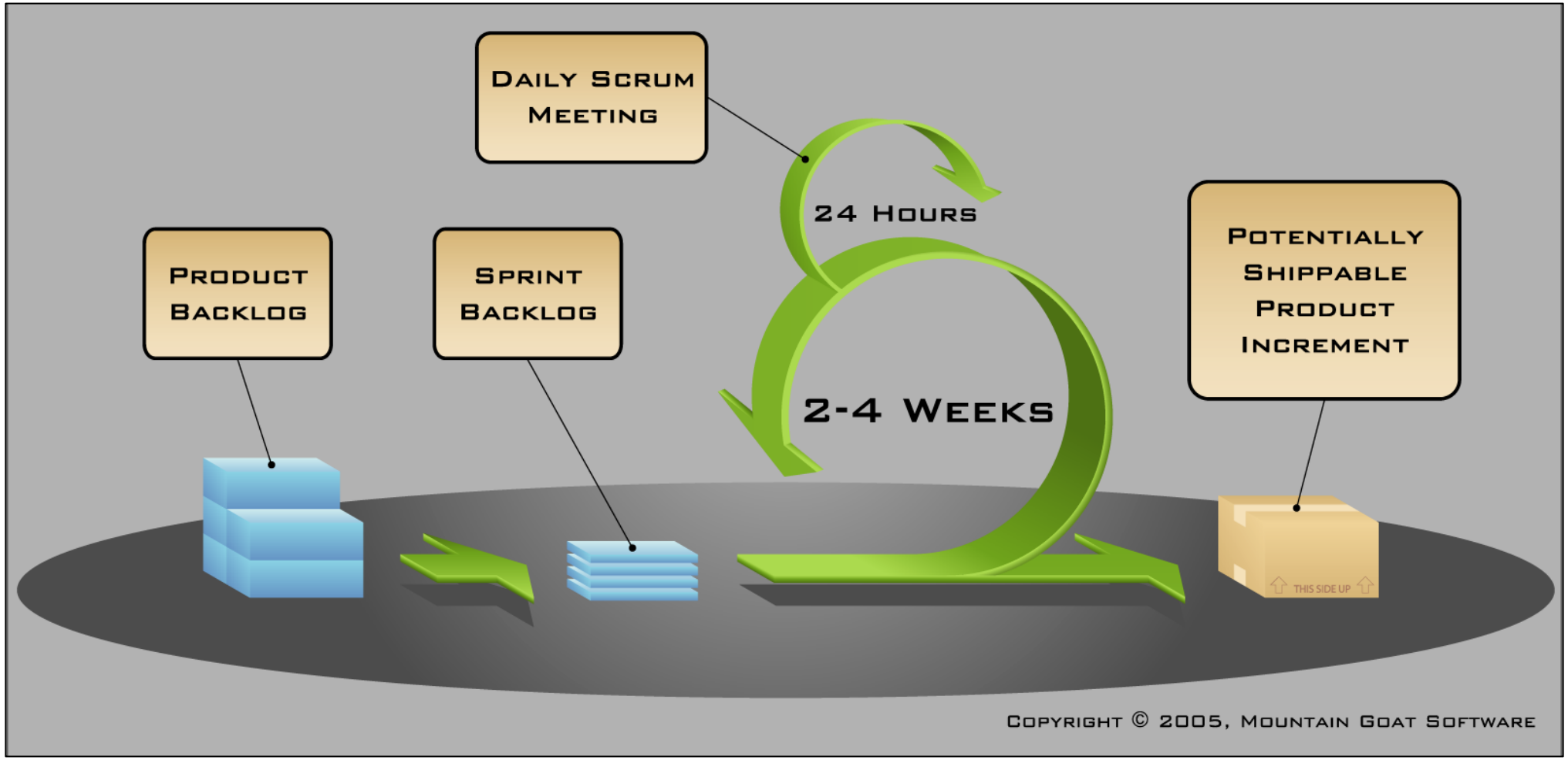
Continuous Integration & Delivery (CI/CD)

GitLab CI + Docker

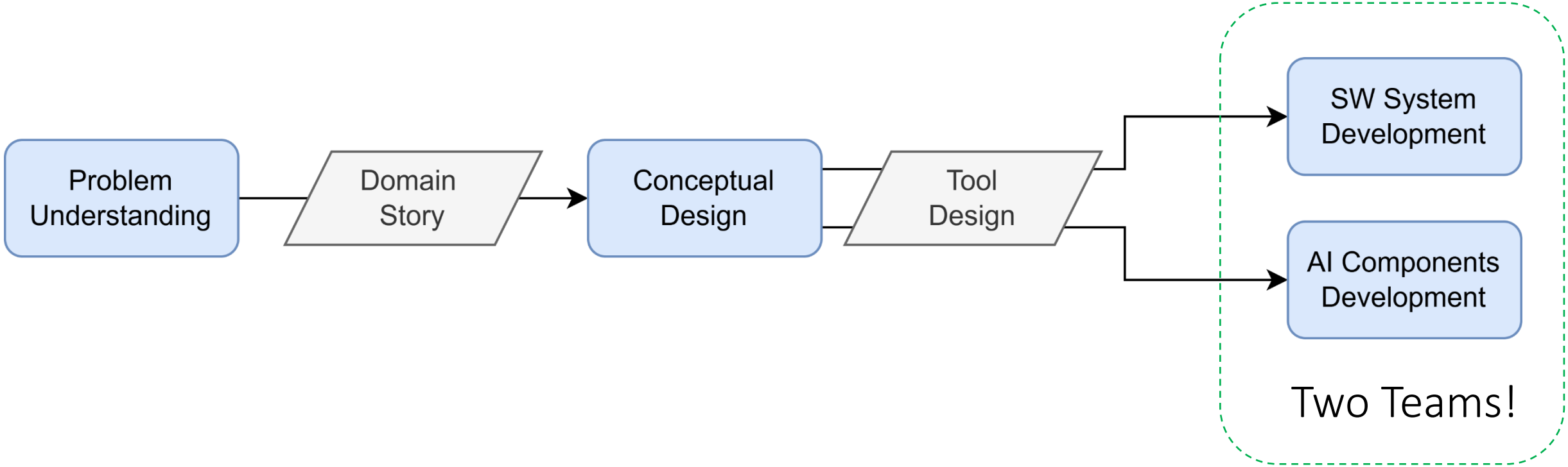
+ whatever helps you!

- Communication
- Documentation
- Programming Lanuages/Frameworks
- Open-Source Tools

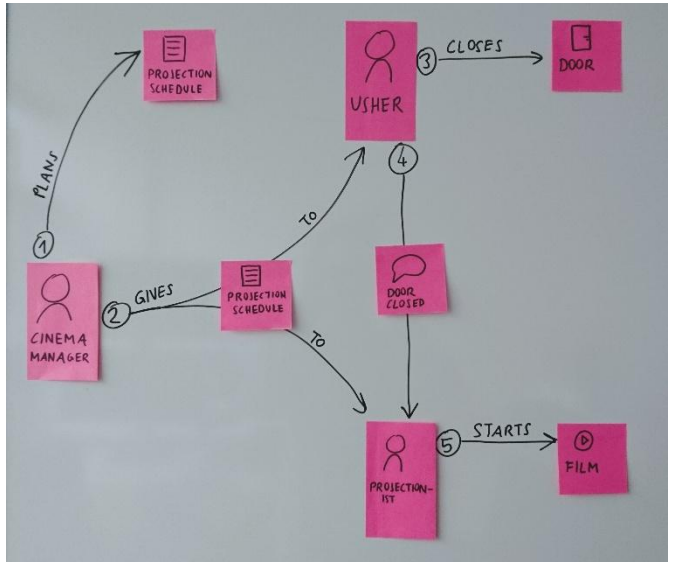
Working in Sprints



Lab Process


Problem Understanding
Domain Story
Conceptual Design
Tool Design
SW System Development
AI Components Development
Two Teams!

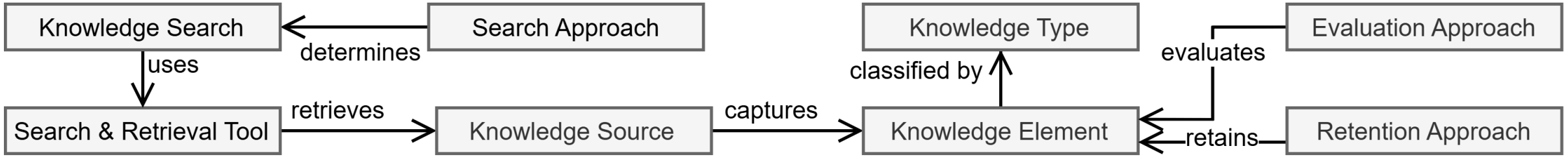
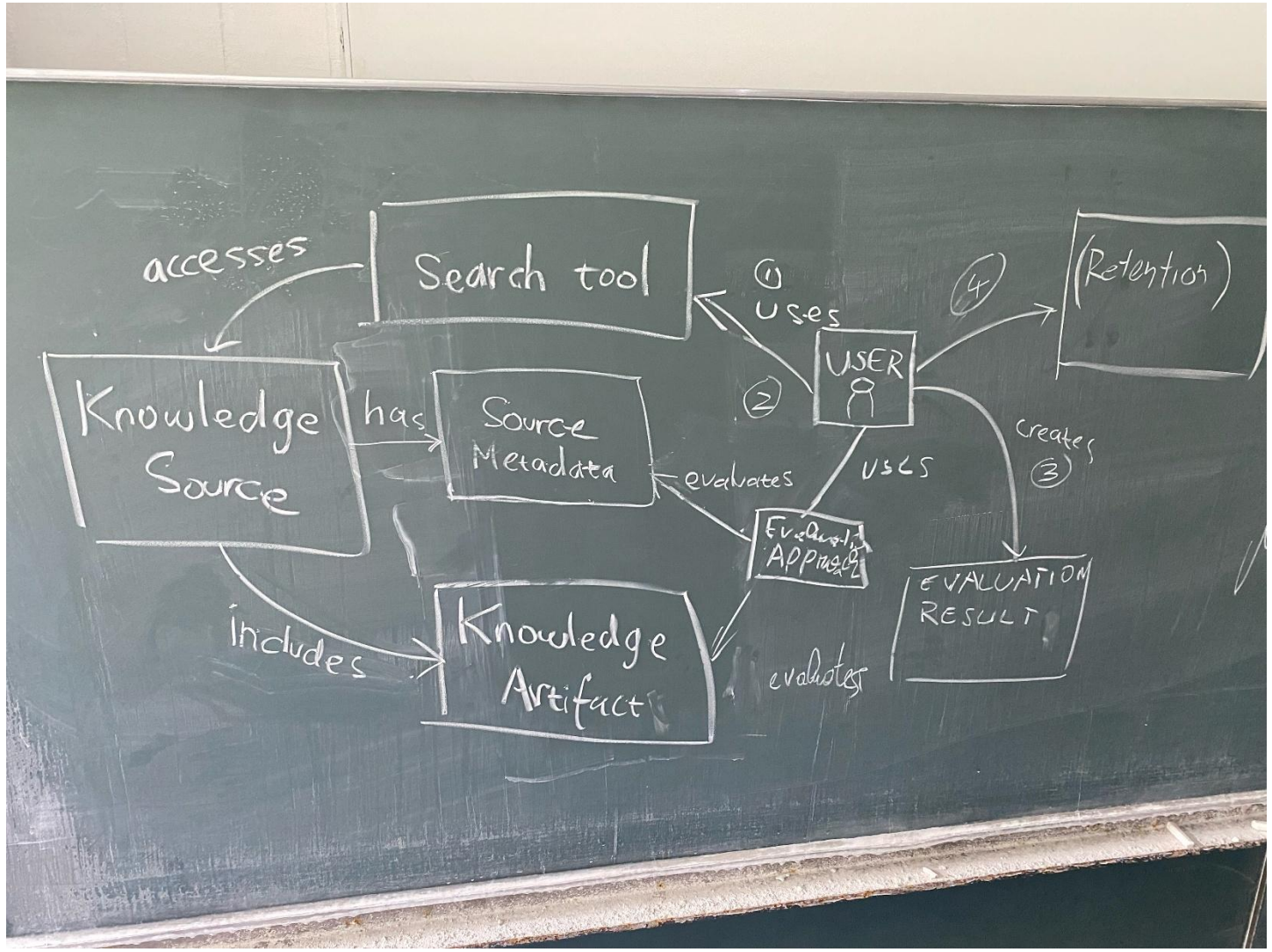
Domain Storytelling



https://domainstorytelling.org/quick-start-guide

- Understand a domain
- Establish a shared language
- Transform domain knowledge into requirements

Reuse Concept Model:



Domain Storytelling